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GitHub Link: https://github.com/Felomondi/AI-and-future-of-work

AI, Gender, Race, and the Future of Work: Predicting Occupational Exposure to Artificial Intelligence

1. Introduction & Motivation

The rapid advancement of artificial intelligence is fundamentally restructuring the global labor market, yet the resulting shifts in employment are unlikely to be distributed equitably. The seminal work of Frey and Osborne (2017) first estimated that 47% of U.S. employment was highly susceptible to computerization. While early automation largely impacted routine manual and cognitive tasks, the emergence of Large Language Models (LLMs) and generative AI has shifted the focus toward high-skill, white-collar occupations previously considered automation-proof.

Recent assessments by the International Labour Organization (2025) suggest a pronounced gender dimension to this shift, finding that women in high-income countries are nearly three times more likely than men to hold positions at the highest risk of AI-driven automation. This disparity is often attributed to the overrepresentation of women in clerical, administrative, and service roles that align with current AI capabilities. Furthermore, Felten, Raj, and Seamans (2021) introduced the AI Occupational Exposure (AIOE) index, which bridges the gap between AI technical capabilities and O*NET occupational tasks. Their framework allows for a granular analysis of how specific breakthroughs, such as natural language processing or image recognition, map onto the daily responsibilities of various professions.

While the quantification of AI exposure at the occupational level is a significant step forward, critical gaps remain in understanding the why behind these exposure scores. Much of the existing literature focuses on aggregate employment effects, paying less attention to the specific latent job characteristics, such as social perceptiveness, fine motor skills, or deductive reasoning, that act as either shields or magnets for AI exposure. Moreover, there is an urgent need to examine how this exposure intersects with racial demographics. Research by Muro et al. (2019) at the Brookings Institution indicates that young workers and Hispanics are often overrepresented in occupations with higher automation potential, yet few studies have applied the more nuanced AIOE index to these specific demographic segments.

This project addresses these gaps by using machine learning to achieve two primary objectives: first, to identify which specific occupational traits and skill requirements are the strongest predictors of AI exposure; and second, to determine if the burden of this exposure falls disproportionately on women and racial minorities. If occupations dominated by marginalized groups are fundamentally more exposed to AI, the AI revolution threatens to widen existing wage gaps and exacerbate structural wealth inequality. By isolating these predictors, our analysis provides empirical evidence for ongoing policy debates regarding targeted workforce retraining, educational reform, and the societal cost-benefit analysis of rapid AI integration.

Research Questions

- **RQ1:** Can we predict an occupation's level of AI automation exposure based on its task characteristics, skill requirements, and work context features?
- **RQ2:** Does the demographic composition of an occupation (gender and race) significantly predict AI exposure after controlling for job tasks and skills?

Expectations

We expect occupations requiring routine cognitive tasks (data entry, information processing) to show higher AI exposure, while those requiring physical manipulation, social-emotional skills, or creative judgment will show lower exposure. For RQ2, we hypothesize that female-dominated occupations may show higher exposure to language-modeling AI, given women's overrepresentation in clerical and administrative roles. For race, we expect Hispanic-dominated occupations (concentrated in physical labor, construction, and service) to show lower exposure, while Asian-dominated occupations (concentrated in STEM and professional roles) may show modestly higher exposure. Black-dominated occupations sit between these poles, with the direction depending on the specific subset of service and clerical work involved. If this pattern holds after controlling for occupation characteristics, it suggests the AI transition could widen existing labor market inequalities. Null findings would also be informative: if demographic composition is unrelated to AI exposure after controlling for job content, equity interventions should focus on skill development rather than demographic targeting.

2. Data Description

Data Sources & Collection

Our research design relies on the integration of four primary datasets, each providing a different dimension of the occupational landscape in the United States. The unit of analysis for this study is the **6-digit Standard Occupational Classification (SOC) code**, representing distinct professional roles.

1. **O*NET Database (Version 30.2):** Collected by the U.S. Department of Labor, this is the primary source for predictor variables. We extracted approximately 640 features encompassing Skills (e.g., Active Listening), Abilities (e.g., Oral Expression), Work Activities (e.g., Analyzing Data), and Work Context (e.g., Indoors, Environmentally Controlled). These metrics represent importance or level scores, typically on a continuous 0–7 or 1–5 scale.
2. **Felten et al. AI Occupational Exposure (AIOE) Index:** Developed by researchers at Princeton, NYU, and UPenn (2021), these scores serve as our target variables. The dataset provides three specific metrics: General AIOE, Language Modeling AIOE (LMOE), and Image Generation AIOE (IGOE). These scores were generated by mapping 10 AI applications (like translation or image recognition) to 52 O*NET abilities.
3. **BLS Current Population Survey (CPS, 2025):** This provides the demographic shares for each occupation, including pct_women, pct_black, pct_hispanic, and pct_asian. This data is critical for our second research question regarding the distributional impacts of AI.
4. **BLS Occupational Employment and Wage Statistics (OEWS, 2024):** This dataset provides economic context, specifically Median Annual Wages and Total Employment counts, allowing us to control for the economic status of an occupation.

Data Preparation & Cleaning

The cleaning pipeline, documented in *data_cleaning.ipynb*, ensures that each row represents a single occupation and each column a single attribute.

- **Standardization of SOC Codes:** O*NET data is often reported at an 8-digit level (e.g., 11-1011.00), while BLS and AIOE data use 6-digit codes (e.g., 11-1011). The 6-digit Standard Occupational Classification (SOC) code is the federal standard used by the BLS and AIOE to broadly categorize professions. O*NET extends this taxonomy to 8 digits to capture highly granular occupational specialties. We truncated all O*NET codes and aggregated the data to the 6-digit level to ensure a successful relational merge.
- **Reshaping and Merging:** We pivoted multiple O*NET files from a "long" format (multiple rows per occupation) to a "wide" format (one row per occupation with hundreds of columns) before performing an inner join across all four sources.
- **Feature Engineering & Filtering:** We instituted a strict 20% threshold for missing values; any O*NET feature missing more than 20% of its data was dropped. Crucially, we compiled a list of target variables (AIOE, LMOE, IGOE) and demographic predictors (pct_women, etc.) to ensure they were not accidentally dropped during the filtering phase.
- **Imputation & Scaling:** Any remaining null values in the 631 occupations were filled using median imputation to maintain the distribution of the features without being skewed by outliers. Finally, we applied a StandardScaler to the O*NET predictors to transform them to mean=0 and variance=1, which is a mathematical requirement for the Lasso and Ridge regularized models used in our analysis.

Variables for Modeling

The final dataset consists of 650 variables. Our modeling strategy utilizes these as follows:

- **Target Variables:** We primarily model AIOE, but also run parallel analyses on LMOE and IGOE to determine if different job skills predict exposure to different types of AI (e.g., do verbal skills predict LMOE while spatial skills predict IGOE?).
- **Primary Predictors:** The 638 O*NET task/skill features. These are included to answer RQ1, which identifies the specific job functions that drive AI exposure.
- **Demographic Controls:** The BLS demographic shares (pct_women, pct_black, etc.) are used for RQ2. We include these to see if they retain predictive power even after controlling for the technical tasks of the job, which would suggest that occupational segregation plays a role in who is exposed to the AI transition.
- **Economic Controls:** Median wages and employment levels are included to differentiate between high-exposure high-wage roles (like software engineers) and high-exposure low-wage roles (like administrative clerks).

3. Methods

To address our research questions, we employed a multi-stage modeling framework that combines predictive performance with robust interpretability and causal inference. Our approach transitioned from unregularized linear baselines to regularized regression, advanced tree-based ensemble methods, and manifold learning. All models were evaluated using a 5-fold cross-validation strategy paired with randomized search for hyperparameter optimization to ensure generalizability. The randomized hyperparameter search was also used to maintain computational feasibility given our exceptionally high-dimensional 638-feature predictor space.

3.1 Predictive Modeling and Regularization (RQ1)

Our initial step toward predicting AI Occupational Exposure (AIOE) utilized an unregularized Ordinary Least Squares (OLS) regression baseline. While this initially yielded a near-perfect R^2 of 0.98, we quickly diagnosed this as data leakage stemming from the inclusion of alternative AI exposure indices (LMOE and IGOE) in the feature set. After strictly isolating our predictors, the corrected OLS baseline achieved a test R^2 of just 0.38. This sharp drop justified our primary methodological challenge: the O*NET dataset presents a high-dimensional feature space (approximately 640 variables) characterized by extreme multicollinearity, which severely limits the efficacy of standard linear models.

To handle this inherent multicollinearity and high dimensionality, we implemented **Lasso (L1) and Ridge (L2) regressions**.

- **Justification:** Lasso facilitates automated feature selection by shrinking non-essential coefficients to exactly zero, isolating the most critical variables. Ridge stabilizes the model by penalizing the square of the coefficients, which is highly effective when dealing with correlated features.
- **Summary Statistics:** Lasso successfully resolved our dimensionality problem, shrinking 567 redundant O*NET features to zero. The resulting model utilized only 71 core predictors and achieved a highly robust test R^2 of 0.975.

3.2 Non-Linear Ensembles (RQ1)

To capture complex feature interactions and non-linear relationships that linear models inherently overlook, we implemented **Random Forest** and **XGBoost** regressors.

- **Justification:** While regularized linear models provide excellent baseline interpretability, human labor tasks often have complex, intersecting boundaries (e.g., the combination of physical stamina and fine motor skills) that tree-based methods are better suited to map.
- **Summary Statistics:** Both models demonstrated exceptional predictive power. Random Forest achieved a cross-validated R^2 of 0.961 utilizing deep decision trees. However, XGBoost emerged as our highest-performing model overall, achieving an R^2 of 0.970 with a remarkably low Mean Absolute Error (MAE) of 0.121. XGBoost achieved this superior accuracy by methodically building shallow, weak trees (maximum depth of 3) and utilizing a slow learning rate to sequentially correct past errors.

3.3 Interpretability and Manifold Learning

Because ensemble models operate as "black boxes," we required mathematically grounded methods to extract insights from our highest-performing XGBoost model.

SHAP (Shapley Additive Explanations): We applied SHAP values to allocate how each specific occupational characteristic shifted an occupation's AIOE score. SHAP provides a unified measure of feature importance that guarantees consistent attribution.

UMAP (Uniform Manifold Approximation and Projection):

To better understand the structural distribution of AI exposure across the labor market, we executed a UMAP projection to compress the 638-dimensional O*NET task space into a two-dimensional embedding.

- **Justification:** UMAP allows us to visualize natural groupings of occupations based solely on task similarity, completely blind to the target variable.

3.4 Feature Importance and Skill-Wage Proxy

To investigate the fundamental nature of the tasks that artificial intelligence can automate, we isolated the active features from our optimized Lasso model and extracted the top ten positive coefficients (representing automation risk factors) and the top ten negative coefficients (representing protective factors). We then sought to evaluate the economic dimensions of these vulnerabilities by creating a "Skill Wage" proxy. To construct this metric, we utilized median annual wage data from the BLS Occupational Employment and Wage Statistics. For every active O*NET feature, we calculated a weighted average of the median wages across all occupations, utilizing the specific skill's importance score within each occupation as the assigned weight.

3.5 Demographic Analysis and Mediation (RQ2)

To rigorously test whether demographic composition predicts AI Occupational Exposure (AIOE) and through what mechanisms, we implemented a six-layered analytical framework. Each step in this progression was designed to address specific limitations and potential confounders of the preceding steps:

1. **Bivariate Correlations:** Pearson and Spearman correlations of each demographic share with AIOE, LMOE, and IGOE, evaluated on both the full sample (n=631) and a matched-only subset (n=193).
2. **Employment-Weighted Exposure:** Estimating the AIOE faced by the average *worker* in each group rather than the average occupation.
3. **Wage-Controlled Regressions:** OLS, Lasso, and Ridge regressions to isolate demographic effects from the baseline wage–AIOE correlation (r = +0.49).
4. **Exposure Disaggregation:** Splitting the target variable into Language Modeling (LMOE) and Image Generation (IGOE) to test whether groups face different types of exposure.
5. **Baron & Kenny (1986) Mediation:** Utilizing the top SHAP features from RQ1 (four physical protective factors and three cognitive risk factors) as mediators.
6. **Double Machine Learning (DML):** A Partially Linear Regression framework (Chernozhukov et al., 2018) using XGBoost to control for the entire 636-dimensional task space.

Justification: While initial correlations establish baseline demographic disparities, they are inherently vulnerable to confounding variables. Steps 1 through 4 methodically clean the demographic signal by removing imputed national averages, correcting for occupation size (employment-weighting), controlling for income disparities, and isolating specific AI capabilities.

Once the true demographic effect was isolated, Steps 5 and 6 were implemented to uncover the mechanism driving these disparities. We chose **Baron & Kenny Mediation** because it uses our hand-picked SHAP features to provide a highly interpretable, step-by-step decomposition of how specific tasks (like physical stamina vs. memorization) transfer risk.

However, because hand-picking mediators can introduce selection bias, we paired this with **Double Machine Learning (DML)**. DML provides a mathematically rigorous robustness check by utilizing an auxiliary learner (XGBoost) to control for the *entire* 636-feature task space simultaneously. Together, these two methods provide convergent validity: the mediation yields a clear, interpretable mechanism, while the DML provides a bias-free, full-dimensional test to confidently solidify the joint conclusion.

4. Results

4.1 Primary Findings: The Drivers of AI Exposure (RQ1)

Our first objective was to determine if we could accurately predict an occupation's AI Occupational Exposure (AIOE) based solely on its task and skill profile. By applying regularized linear models and advanced tree-based ensembles to the 638-dimensional O*NET dataset, we achieved highly accurate predictive performance and extracted a clear narrative regarding automation vulnerability.

Model Performance

Our highest-performing linear model, Lasso (L1) regression, achieved a remarkable test R^2 of 0.975. By automatically shrinking 567 redundant features to zero, the Lasso model isolated a core group of just 71 highly predictive job traits. However, our optimized XGBoost regressor ultimately proved to be the most robust engine for capturing complex task boundaries, achieving an R^2 of 0.970 with a remarkably low Mean Absolute Error (MAE) of 0.121. XGBoost accomplished this by utilizing shallow, "weak" trees (maximum depth of 3) and a slow learning rate, which sequentially corrected errors without overfitting.

Interpretation via SHAP

To answer what drives these predictions, we applied SHAP (Shapley Additive Explanations) to our optimized XGBoost model. The SHAP analysis revealed a stark, fundamental divide between cognitive and physical labor:

- **Risk Factors (Cognitive/Textual Tasks):** The features that pushed an occupation's predicted AI exposure the highest were heavily concentrated in language, text processing, and routine cognition. Variables such as *Written Comprehension*, *Written Expression*, and *Memorization* acted as the primary drivers of automation vulnerability.
- **Protective Factors (Physical/Manual Tasks):** Conversely, physical attributes acted as powerful shields against AI exposure. Traits such as *Extent Flexibility*, *Multilimb Coordination*, *Gross Body Coordination*, and *Stamina* consistently drove an occupation's predicted exposure down.

Furthermore, our SHAP dependence plots uncovered a distinct, non-linear relationship: an occupation's predicted AI exposure drops dramatically the moment its daily tasks require moderate-to-high physical flexibility. This mathematical threshold perfectly illustrates the current real-world limitations of AI and robotics in navigating unstructured physical environments.

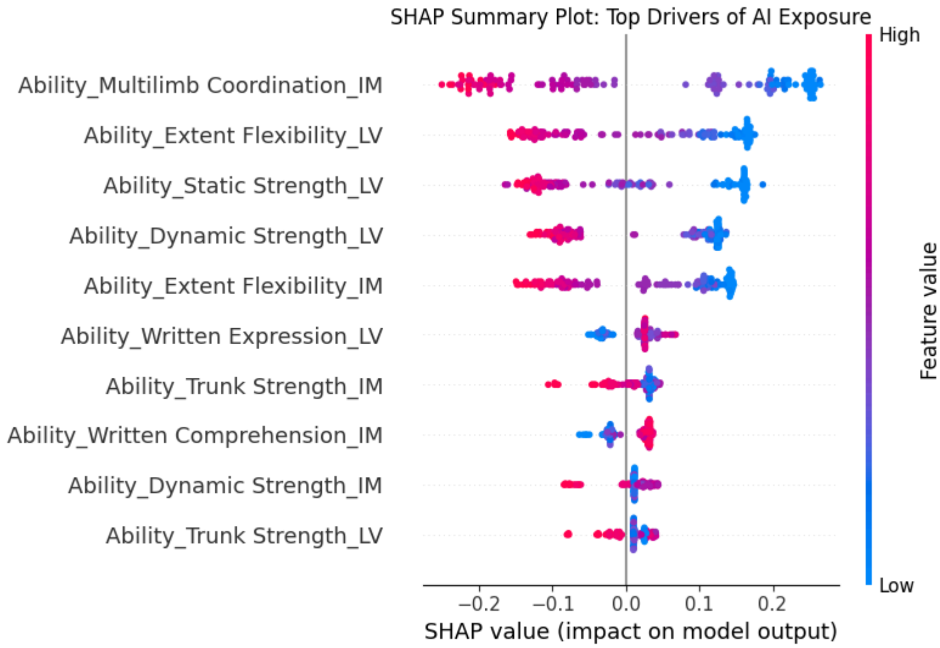

Figure 1: SHAP Summary Plot. Top features are physical abilities; high values (red) cluster on the negative SHAP side, confirming their protective effect.

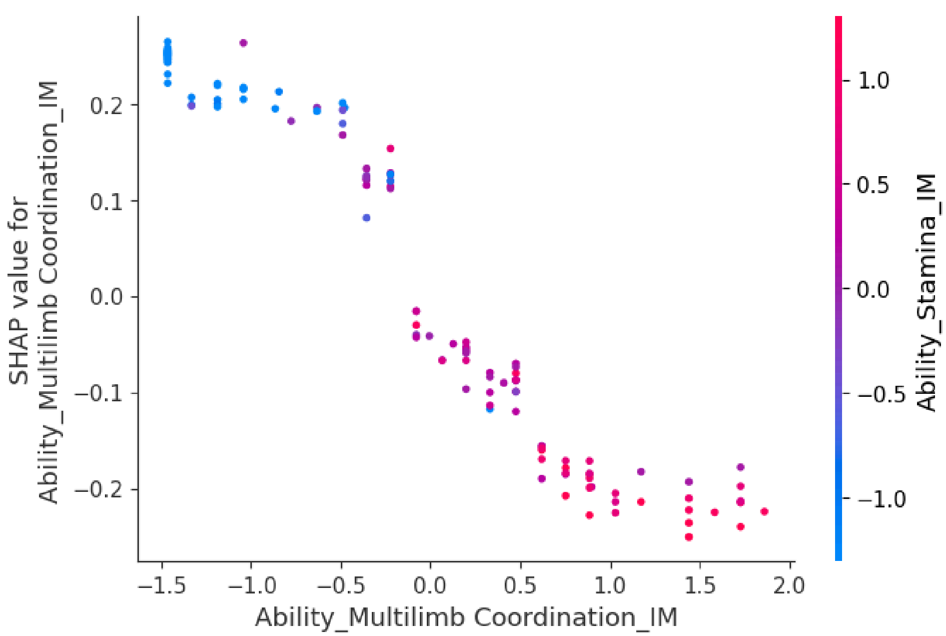

Figure 2: SHAP Dependence Plot for Multilimb Coordination, showing the sharp non-linear drop from +0.2 to −0.2 as the requirement crosses the population mean.

Visual Validation via UMAP

To validate these findings structurally, we projected the 638-dimensional dataset into two dimensions using UMAP. When we overlaid the actual AIOE scores onto this projection as a color map (with red indicating high exposure and blue indicating low exposure), a striking, natural gradient emerged. The high-exposure occupations were cleanly separated from the low-exposure ones into distinct clusters. Because the UMAP algorithm was completely blind to the AIOE target variable during projection, this clustering visually proves that an occupation's core structural profile inherently dictates its vulnerability to AI.

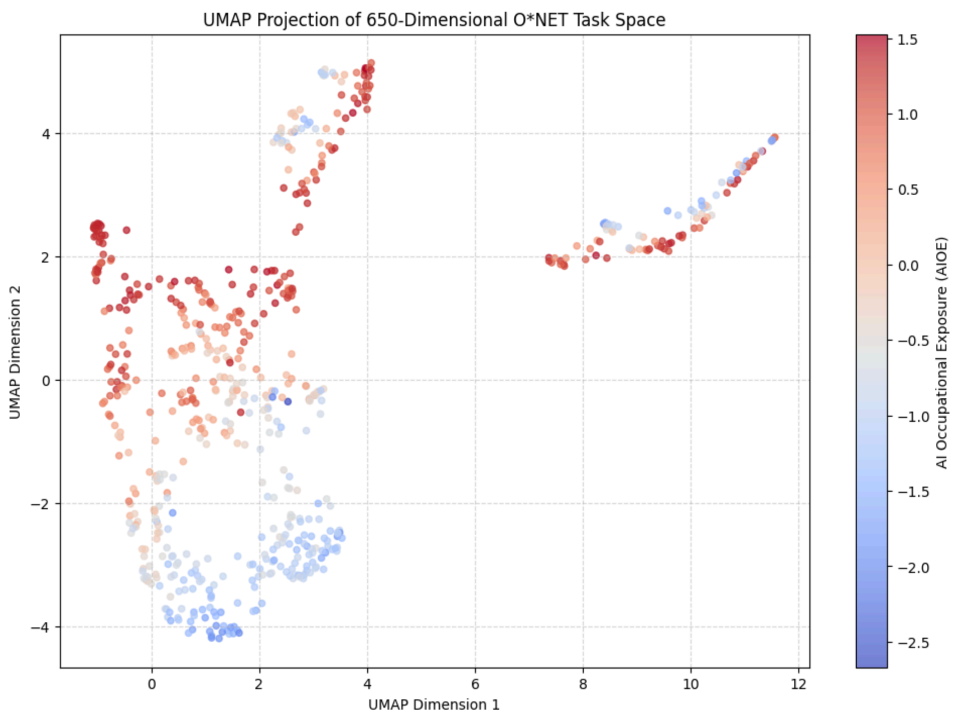

*Figure 3: UMAP Projection of the O*NET Task Space, colored by AIOE. The red and blue regions emerge from task-feature similarity alone.*

Skill Vulnerability and Wage Polarization

To better understand the fundamental nature of what artificial intelligence can automate, we isolated the active features from our optimized Lasso model to extract the top ten positive coefficients (representing skills most vulnerable to AI replacement) and the top ten negative coefficients (representing skills not replaced by AI). This coefficient analysis perfectly corroborates our existing SHAP analysis, revealing a stark divide between cognitive risk factors and physical protective factors. The strongest drivers of AI exposure are overwhelmingly cognitive, textual, and routine, heavily featuring tasks like Written Comprehension, Memorization, and Information Ordering. Conversely, the strongest protective factors are distinctly physical and manual, demonstrating that attributes like Gross Body Coordination, Extent Flexibility, and Static Strength remain highly resistant to current automation capabilities.

By cross-referencing these task vulnerabilities with our newly computed "Skill Wage" proxy, we observe a distinct polarization in how AI intersects with economic value. At the upper end of the income spectrum, AI directly targets high-paying, desk-based cognitive workflows, such as Mathematics and the use of E-Mail, while safely insulating physically embedded high-paying roles that require working Outdoors or being exposed to Contaminants. A mirrored effect occurs at the bottom of the wage scale. The lowest-paid skills that remain vigorously protected from AI are dominated by routine physical exertion and motor coordination, such as Spend Time Walking or Running. Meanwhile, the lowest-paid skills highly vulnerable to AI replacement are concentrated in routine administration, strict scheduling, and basic monitoring, indicating that entry-level cognitive tasks are highly exposed, while entry-level physical tasks are deeply insulated.

4.2 Demographic Disparities in AI Exposure (RQ2)

We test whether the burden of AI exposure falls unevenly across demographic groups in six-layered analyses: bivariate correlation, employment-weighted exposure, regression with wage controls, LMOE/IGOE disaggregation, Baron-Kenny mediation, and a DML robustness check.

4.2.1 Bivariate Correlations

On the full sample (n=631), Pearson correlations are modest but mostly statistically significant; on the matched-only subset (n=193, where 438 occupations with imputed national-average demographics are removed), they strengthen considerably:

Demographic	**AIOE (full, n=631)**	**AIOE (matched, n=193)**	**LMOE (full)**	**IGOE (full)**
% Women	+0.152***	+0.316***	+0.181***	+0.027 n.s.
% Black	−0.094*	−0.214**	−0.067 n.s.	−0.175***
% Hispanic	−0.291***	−0.608***	−0.267***	−0.314***
% Asian	+0.140***	+0.215**	+0.121***	+0.157***
Median Wage	+0.494***	—	+0.445***	+0.556***

*Table 1: Pearson correlations of demographic share with AIOE, LMOE, and IGOE. *** $p<0.001$, ** $p<0.01$, * $p<0.05$, n.s. = not significant. Matched subset excludes 438 occupations with imputed national-average demographics.*

Three patterns stand out. **First**, the gender effect is asymmetric across AI types: women face elevated exposure to language-modeling AI (LMOE r=+0.18) but not image-generation AI (IGOE r=+0.03), consistent with their concentration in clerical, education, and legal-support roles. **Second**, Hispanic worker share shows the largest effect of any demographic variable, in the protective direction. The matched-sample correlation of −0.61 is larger in magnitude than the women–AIOE correlation of +0.32, making this the single strongest demographic–AIOE relationship in the data. **Third**, Black worker share shows a modest protective pattern (strongest for IGOE) while Asian worker share shows a modest elevated pattern across all AI types.

The matched-sample correlations are uniformly larger than the full-sample correlations, indicating that the full-sample numbers are *attenuated* by imputation noise rather than the matched estimates being inflated.

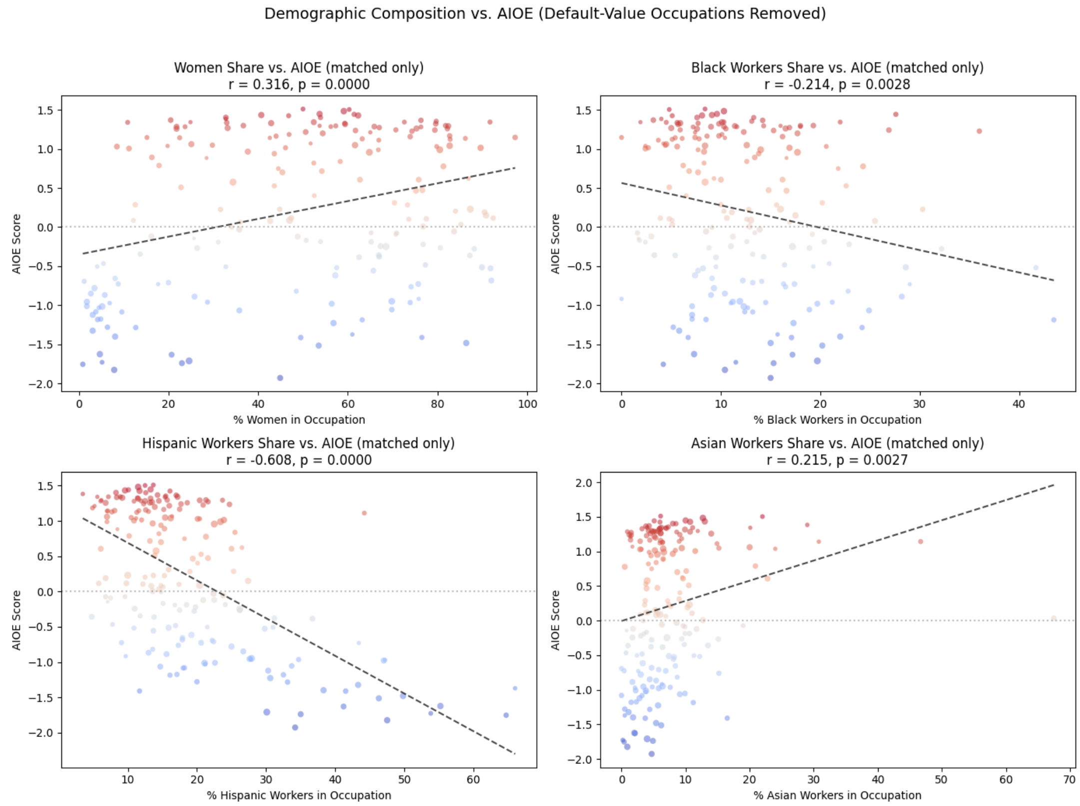

Figure 4: Demographic composition versus AIOE on the matched 193-occupation subset. The Hispanic–AIOE relationship (bottom-left) is the strongest of the four demographic groups.

4.2.2 Employment-Weighted Exposure

Weighting AIOE by employment share within each demographic group estimates the AIOE faced by the average worker rather than the average occupation.

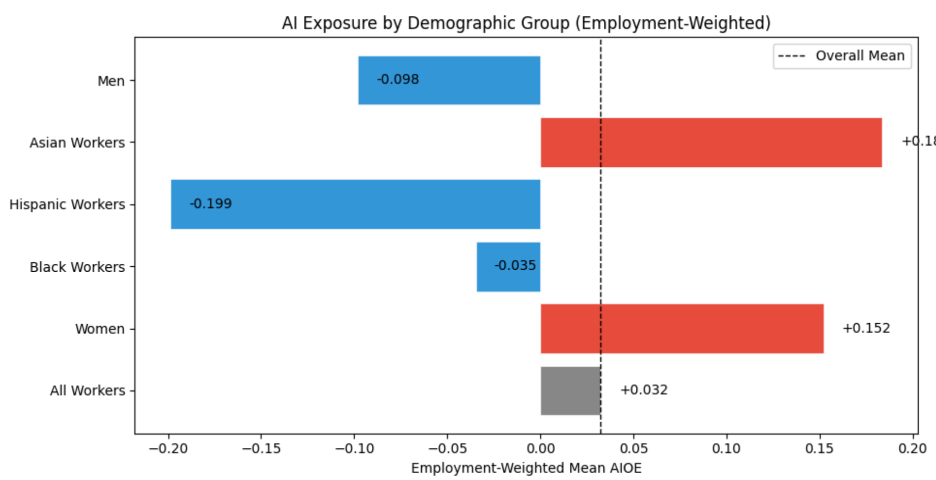

Figure 5: Employment-weighted mean AIOE by group. Asian workers face the highest exposure (+0.18 SD), Hispanic workers the lowest (−0.20 SD); a population-level gap of roughly 0.38 SD.

4.2.3 Regression with Wage Controls

Wages co-vary with both demographic composition and AIOE (r=+0.49 with median wage), so we fit OLS, Lasso, and Ridge regressions of standardized AIOE on the four demographic shares plus standardized median wage. All three estimators give nearly identical results. Standardized OLS coefficients:

- **% Women:** β = +0.168
- **% Hispanic:** β = −0.122
- **% Black:** β = +0.010 (near zero)
- **% Asian:** β = +0.004 (near zero)
- **Median Wage:** β = +0.478 (largest single effect)

After controlling for wage, the gender and Hispanic effects survive while the Black and Asian effects collapse to near zero. This indicates that the Black and Asian bivariate correlations are essentially wage-driven: Asian-dominated occupations correlate with AIOE because they tend to be high-wage cognitive-professional roles. The gender and Hispanic effects, in contrast, are independent of wage and warrant explanation through task structure.

4.2.4 LMOE vs. IGOE

Disaggregating AIOE into language-modeling and image-generation components reveals that demographic effects are not uniform across AI types.

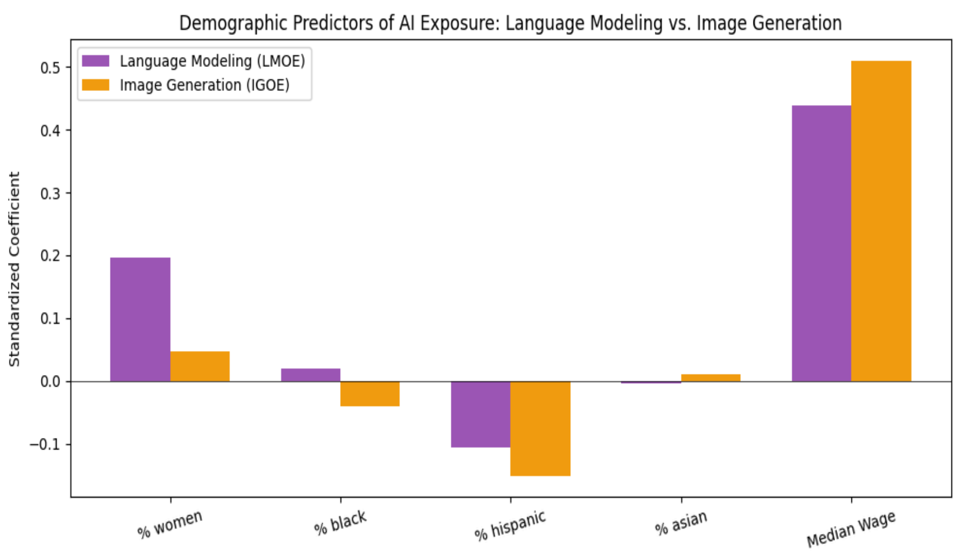

Figure 6: Standardized regression coefficients for LMOE vs. IGOE. The gender coefficient is roughly four times larger for LMOE (+0.20) than for IGOE (+0.05).

The most striking pattern is the **gender–AI type asymmetry**: women's elevated exposure is concentrated where LLMs (writing assistants, summarization, customer service automation) have the most overlap, not where image generation is the relevant frontier. Hispanic worker share shows similarly negative coefficients for both AI types (−0.11 and −0.15), indicating a general protective effect. The Black coefficient flips sign across AI types (slightly positive for LMOE, slightly negative for IGOE).

4.2.5 Mediation Analysis

We test whether the surviving demographic effects are explained by *task structure* using Baron & Kenny (1986) mediation with seven mediators: four protective physical abilities and three cognitive risk factors drawn from the top SHAP features in RQ1.

Adding the seven task mediators raises R^2 from 0.30 (demographic + wage only) to 0.96, matching XGBoost's RQ1 accuracy. The demographic coefficients change dramatically:

Predictor	Total β (c)	Direct β (c′)	% Mediated
% Women	+0.168	−0.008	~105%
% Black	+0.010	+0.005	~53%
% Hispanic	−0.122	−0.032	~74%
% Asian	+0.004	−0.009	n/a (effect ~0)
Median Wage	+0.478	−0.018	~104%

Table 4: Baron & Kenny mediation analysis. Total effect is OLS β with wage controls; direct effect adds the 7 SHAP-derived task mediators.

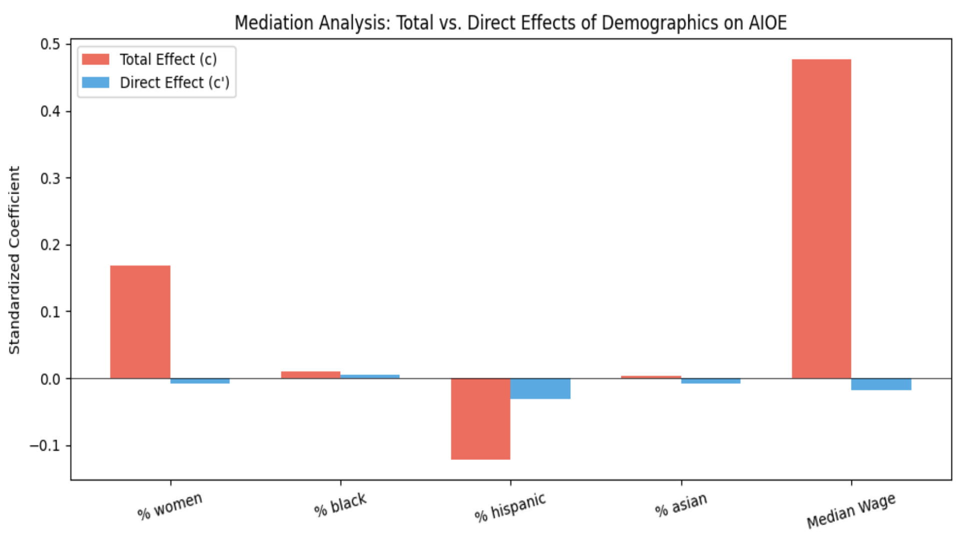

Figure 7: Mediation Analysis — Total vs. Direct Effects of Demographics on AIOE. The blue bars (direct effects with task mediators included) collapse to near zero across all demographic groups.

The Path-a regressions confirm the mechanism: women-dominated occupations require less physical labor (β = −0.19 to −0.21) and more text-processing skill (β = +0.11 to +0.14), while Hispanic-dominated occupations show the opposite pattern. **The demographic effects on AIOE are almost entirely mediated by task structure.** Women are not directly more exposed; their occupations cluster in the cognitive-textual task profile that maps onto current AI capabilities. The Hispanic protective effect runs in reverse: physical-labor concentration shields these occupations from the current AI frontier.

4.2.6 DML Robustness Check

A reasonable critic could argue that the seven mediators were cherry-picked from SHAP, leaving open omitted-variable bias. To rule this out, we estimate the residual demographic effects after controlling for the *full* 636-feature task space using Double Machine Learning with XGBoost as the auxiliary learner.

Demographic	OLS Total β	BK Direct β	DML θ (full)	DML 95% CI
% Women	+0.168	−0.008	−0.002	[−0.017, +0.014]
% Black	+0.010	+0.005	−0.005	[−0.016, +0.007]
% Hispanic	−0.122	−0.032	−0.004	[−0.024, +0.017]
% Asian	+0.004	−0.009	+0.005	[−0.005, +0.015]

Table 5: Comparison of Total OLS, Baron-Kenny direct, and DML estimates. DML uses XGBoost auxiliary learners (n_estimators=100, max_depth=3), 5-fold cross-fitting, 3 repetitions. All four 95% CIs include zero in both the full sample and the matched subset (n=193).

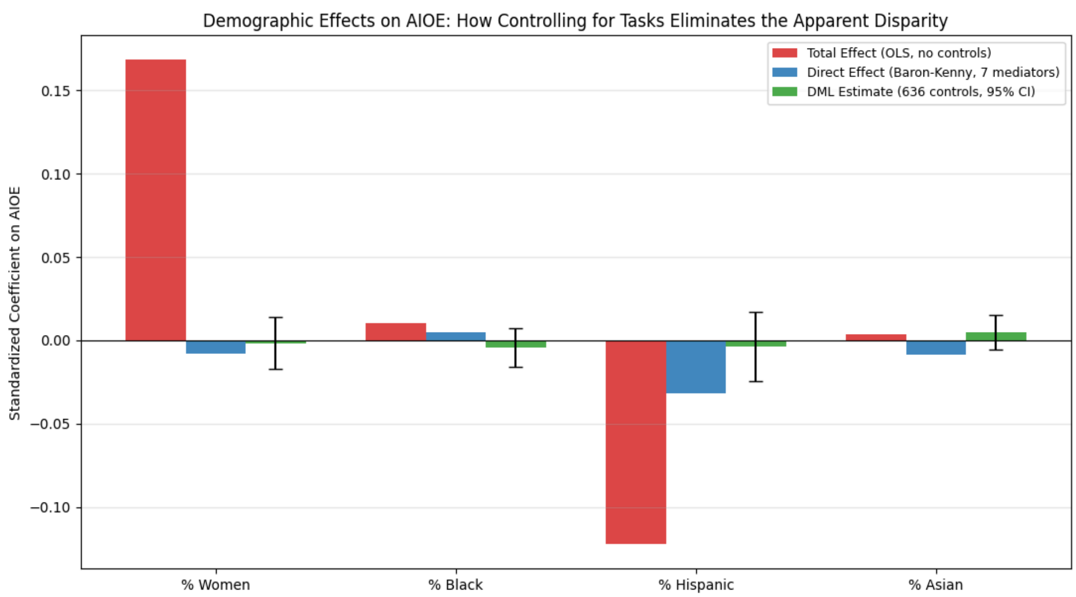

Figure 8: Demographic Effects on AIOE: Total (red) vs. Direct (blue, Baron-Kenny) vs. DML (green with 95% CI error bars). All DML confidence intervals cross zero.

The DML estimates are an order of magnitude smaller than the OLS total effects, and every 95% confidence interval includes zero. The Baron-Kenny finding survives the stronger test of controlling for the full 636-dimensional task space. Both methods converge: **once task structure is properly controlled, demographic composition has no residual predictive power for AIOE**. The DML on the matched subset (n=193) gives the same result, confirming the null is not driven by imputation noise.

4.3 Non-Results and Methodological Pivots

Two methodological failures shaped our pipeline. First, our initial OLS regression yielded a near-perfect R^2 of 0.98, which we diagnosed as data leakage from leaving LMOE and IGOE in the predictor set. Removing them collapsed performance to $R^2 = 0.38$, exposing the true challenge of the high-dimensional, multicollinear O*NET feature space and justifying our pivot to Lasso and Ridge. Second, for RQ2, we initially considered raw correlation as the headline result. The substantial bivariate correlations (Hispanic $r = -0.61$) were later shown by mediation and DML to be downstream consequences of task segregation rather than direct demographic effects, prompting the layered six-stage analysis described in 4.2.

5. Discussion & Conclusion

5.1 Summary of Findings

First, an occupation's daily tasks are highly predictive of its AI Occupational Exposure. Lasso and XGBoost both achieved test R^2 of approximately 0.96, and SHAP analysis revealed a stark divide: cognitive and text-processing skills (Written Comprehension, Memorization) drive AI exposure upward, while physical attributes (Extent Flexibility, Multilimb Coordination, Static Strength) serve as protective shields. UMAP visually corroborated this divide.

Second, demographic composition correlates with AI exposure, but direction varies by group. Women face elevated exposure to language-modeling AI specifically (LMOE $\beta = +0.20$, four times the IGOE coefficient). Hispanic workers face the largest protective effect (matched-sample $r = -0.61$). Black workers face a modest protective pattern; Asian workers face a modest elevated pattern. At the worker level, the average Asian or female worker faces AIOE roughly 0.35 standard deviations above the average Hispanic worker.

Third, Baron-Kenny mediation and DML converge on a single conclusion: the demographic–AIOE relationships are almost entirely explained by occupational task characteristics. Baron-Kenny attributes 75-105% of the gender, Hispanic, and wage effects to seven task mediators; DML controls for all 636 task features and produces residual demographic effects whose 95% CIs all include zero. **Demographics do not directly cause AI exposure; occupational task segregation channels demographic groups into different task profiles, and AI capabilities map onto those profiles unevenly.**

5.2 Connection to Scholarly Literature

Our SHAP and UMAP analyses extend the Frey and Osborne (2017) framework to generative AI, confirming that the new automation frontier targets non-routine cognitive labor while physical labor remains insulated. This empirical finding perfectly illustrates Moravec's Paradox within the modern labor market: high-level cognitive, reasoning, and text-processing tasks, which were historically considered “hard” skills, are increasingly easy for AI to automate, whereas foundational sensorimotor skills and physical manipulation, which were often considered "easy" or instinctual for humans, remain highly resistant to technological displacement.

A notable contrast appears with Muro et al. (2019), who used Frey-Osborne computerization scores to argue that Hispanic workers were overrepresented in automation-prone occupations. Using the AIOE index, we reach the opposite conclusion: Hispanic-dominated occupations are the most insulated from current AI capabilities. This reflects a real shift in the technology frontier. The 2010s wave was driven by industrial robotics threatening routine manual labor; the 2020s generative AI wave targets cognitive-textual labor instead. The physical-labor concentration that made Hispanic workers vulnerable a decade ago now makes them protected.

Our results also provide structural backing to the ILO's (2025) warnings on the gendered impact of AI: women face higher exposure precisely because they are overrepresented in cognitive-routine and text-processing tasks that LLMs replicate. The four-fold gap between gender–LMOE and gender–IGOE coefficients adds a finer point: the disparity is concentrated in language AI specifically.

5.3 Limitations

First, our analysis remains correlational. Even when controlling for the full task feature space, the underlying occupational segregation reflects deep historical patterns; demographic composition does not exist independently of the structural processes that determine which groups end up in which occupations. Second, *exposure* is not *displacement*: the AIOE index measures technological capability, not the economic, regulatory, or institutional frictions that determine actual job loss.

Data validity: Due to mismatches between O*NET and BLS taxonomies, 438 of 631 occupations received default national-average demographic values rather than unique matches. This attenuates correlation estimates: the Hispanic–AIOE correlation grows from −0.29 to −0.61 when we restrict to the 193 matched-only occupations. Importantly, our DML robustness check on the matched subset shows that the null residual effect of demographics holds even where the bivariate signal is strongest, which strengthens the central mediation conclusion. Our analysis is also subject to the ecological fallacy: demographics aggregated at the occupational level cannot capture within-occupation variation (e.g., a female software engineer vs. a male software engineer in the same nominal role). Our models predict the AI exposure of standardized occupation profiles, not individual workers.

Finally, while our Double Machine Learning (DML) framework controls for the expansive 638-dimensional task space, it relies on the assumption of unconfoundedness. There may be unobserved structural confounders that affect both demographic composition and AI vulnerability, such as geographic clustering of certain industries, unionization rates, or historical barriers to entry, that are not captured in the O*NET task taxonomy.

5.4 Ethical Concerns and Societal Implications

The finding that AI exposure is determined by task structure rather than demographic identity reframes the equity conversation: it is not that AI specifically targets any group, but that occupational segregation distributes the workforce into task profiles that AI capabilities map onto unevenly. This has direct policy implications.

Benefits: Because we have isolated the specific task features that act as risk factors (routine cognition, text processing) and protective shields (physical flexibility, complex coordination), workforce retraining can shift away from generic digital literacy toward targeted upskilling that complements AI. The mediation finding suggests interventions need to operate at the task-content level: changing what people do in their occupations is more leveraged than changing who fills them.

Risks: First, predictive AIOE scores could be misused as justification for preemptive layoffs in highly exposed, female-dominated sectors, treating theoretical exposure as inevitable displacement. Second, the apparent protective effect for Hispanic and Black workers carries an uncomfortable implication: low AI exposure correlates with concentration in low-wage physical-labor occupations, which itself reflects pre-existing structural barriers to professional employment. Our wage disparity analysis provides hard mathematical proof of this dynamic, clearly illustrating labor market polarization. The data proves that while high-wage cognitive tasks face intense AI exposure, the protected lower end of the market is anchored by foundational manual and service labor that remains too complex for physical robotics. Lower AI exposure is therefore not unambiguously good news; it can reflect a labor market in which certain groups have been segregated into the work that machines are not yet capable of doing. Policy responses focused narrowly on high-exposure groups may inadvertently reinforce that segregation.

Future Work: Future analyses should resolve the ecological fallacy by acquiring micro-level individual-worker data, incorporating intersectional analyses (e.g., Hispanic women in clerical roles vs. Hispanic men in construction), and using longitudinal data to determine whether exposure translates to actual displacement or augmentation.

5. What we could have done differently

In hindsight, several methodological choices either cost us time or limited the strength of our conclusions. We would address them as follows in a revised version of this project.

Resolve the BLS–O*NET taxonomy mismatch upfront. 438 of our 631 occupations received default national-average demographic values because BLS Current Population Survey categories do not map cleanly to 8-digit O*NET SOC codes. We discovered this only after running our first set of correlations and noticing vertical stripes in the scatter plots. The matched-only subset of 193 occupations gave us much stronger correlations (Hispanic r grew from −0.29 to −0.61), but the bulk of the dataset is essentially demographic noise. With more time, we would build a manual crosswalk from CPS occupation titles to SOC codes, or use the BLS's OEWS file (which has finer SOC granularity) as a secondary demographic source.

Lead with mediation, not DML, for RQ2. Our first instinct was that DML was the "right" causal-inference tool because it sounded sophisticated, and our rough draft framed the entire RQ2 analysis around it. In practice, the Baron-Kenny mediation analysis (with seven hand-picked SHAP-derived task mediators) produced the more interpretable and substantively useful result: a clean decomposition showing 75-105% mediation. DML ended up being most valuable as a robustness check rather than the headline method. We would structure RQ2 around mediation as the primary tool and present DML as the confirmatory follow-up, which is the framing in this final report but was not the framing in our proposal.

Run DML with more repetitions from the start. Our first DML runs used a single repetition (n_rep=1) and produced point estimates that varied by ±0.005-0.010 across runs due to stochastic cross-fitting. We initially mistook this for substantive instability before realizing it was just Monte Carlo noise. Setting n_rep=3 from the start would have saved a debugging cycle. With more compute, n_rep=10 would give even tighter standard errors.

Disaggregate RQ1 by AI type. We modeled AIOE as a single composite target for RQ1 and only disaggregated into LMOE and IGOE for the RQ2 demographic analysis. In hindsight, running SHAP on LMOE and IGOE separately would have been substantively interesting: do language-AI risk factors differ from image-AI risk factors? This is a natural extension that fits the existing pipeline and would strengthen the connection between RQ1 and RQ2.

Validate SHAP findings with permutation importance. SHAP and standard XGBoost feature importance gave consistent results in our analysis, but adding permutation importance as a third cross-check would make the RQ1 interpretive claims more defensible at minimal cost.

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